

Cem Ozcelik

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PROFESSIONAL SUMMARY

Senior Data Scientist with an Industrial Engineering background and nearly 5 years of progressive experience owning the **full ML lifecycle** - EDA, feature engineering, model selection, hyperparameter tuning, evaluation, and Docker-based production deployment. Shipped the retrieval foundation of **RAG / LLM systems** in production: Qdrant vector database, SentenceTransformers, and custom-trained Word2Vec on 800K+ document corpora with A/B testing between embedding approaches. Deep stack across Python, advanced SQL, scikit-learn, XGBoost, PyTorch/TensorFlow, FastAPI, and PySpark/Databricks. Industrial Engineering roots provide a native cost-function lens (FP vs FN as business cost) -- directly applicable to insurance pricing, fraud-claim detection, and customer-risk modelling, as demonstrated in an end-to-end Credit Risk pipeline (Lending Club, XGBoost, temporal holdout). Comfortable communicating model outputs and storytelling to C-level and business stakeholders in Turkish and English. Demo: github.com/ccemozclk/amazon-ai-search-engine | github.com/ccemozclk/Credit-Risk

CORE COMPETENCIES

Skills: End-to-End ML Lifecycle, LLM & RAG Pipelines, Vector Databases, MLOps & Model Monitoring, Statistical Modelling, Hyperparameter Tuning (Optuna), Insurance / Risk Modelling Analog, C-Level Storytelling (TR/EN)

Tools: Python (Advanced), SQL (Advanced), scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Hugging Face, SentenceTransformers, Qdrant (Vector DB), FastAPI, Docker, Optuna, PySpark, Streamlit, Git

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effeffe Jan 2026 -- Present

Senior Database Analyst | Maltepe, Istanbul

- Took full ownership of an **end-to-end ML platform**: scaled an internal Django + Docker application into a centralized hub serving **versioned REST APIs for production ML models**, establishing the foundation for monitoring, drift detection, and automated retraining.
- Conducted EDA and statistical analysis to support product decisions, and translated SKU-level Association Rule Mining outputs into cross-sell strategy that drove continuous revenue growth.
- Directed predictive supply chain analytics using Python and advanced SQL, sustaining an **8-hour weekly reduction in operational overtime** through ML-driven forecasting.
- Worked collaboratively with product, growth, engineering, and operations stakeholders -- communicating model outputs to both technical and non-technical audiences.
- **Tech:** Python, SQL, Django, REST APIs, ML pipelines, Docker, PostgreSQL, Association Rule Mining

Anadolu Pet Akvaryum & Effeffe Sept 2024 -- Dec 2025

IT & Data Analytics Specialist | Maltepe, Istanbul

- Architected a full-stack Django web application from scratch (SOLID principles), establishing the foundation that later scaled into the production ML deployment platform.
- Engineered predictive supply chain models in Python and PL/SQL that **reduced stockouts and improved inventory turnover**.
- Conducted Route-Level Profitability Analysis with advanced SQL, delivering a **3% improvement in overall logistics efficiency**.
- **Tech:** Python, Django, PL/SQL, Oracle, SOLID, full-stack deployment

Birinci Automotive

Jan 2022 -- Sept 2024

Mid-Level Data Analyst | Cayirova, Kocaeli

- Developed a **Predictive Maintenance & Anomaly Detection framework** using unsupervised clustering on machine-sensor time-series -- the same statistical primitives that drive fraud-claim detection and behavioural anomaly modelling in insurance.
- Delivered a **15% weekly increase in critical machine availability** and a **15% OEE improvement** through cost-function-aware alerting (downtime cost >> false-alarm cost).
- Engineered robust ETL pipelines in MSSQL and Python for high-volume sensor data, replacing manual reporting with reliable analytical infrastructure.
- **Tech:** Python, MSSQL, ETL, unsupervised clustering, time-series analysis, anomaly detection

Birinci Automotive

June 2021 -- Jan 2022

Junior Data Analyst | Cayirova, Kocaeli

- Conducted EDA and applied statistical methods to workforce datasets, uncovering hidden patterns to optimize HR policies.
- Designed automated data workflows and ingestion pipelines, **improving reporting efficiency by 15%**.
- **Tech:** Python, SQL, statistical analysis, ETL

PROJECTS

E-Commerce AI Search Engine RAG / Vector DB / LLM Foundation

Production-ready microservices architecture serving as the **retrieval foundation for RAG-style systems**. Dual-engine A/B test of SentenceTransformers (all-MiniLM-L6-v2) vs custom-trained Word2Vec on **800K+ product descriptions**, stored in a **Qdrant vector database** (same API surface as Pinecone/Milvus). Custom **LSTM next-token prediction** in PyTorch for intelligent search auto-complete. **Item2Vec collaborative filtering** for "Frequently Bought Together" recommendations. Sub-millisecond cached responses via Redis. Architecture transfers 1:1 to insurance RAG: policy Q&A, claims FAQ chatbot, quote-comparison assistant.

PyTorch, FastAPI, SentenceTransformers, Hugging Face, Qdrant (Vector DB), Redis, PySpark, Docker | github.com/ccemozclk/amazon-ai-search-engine

End-to-End Credit Risk Prediction MLOps & Insurance-Risk Analog

Full lifecycle credit risk pipeline on Lending Club data -- structurally identical to insurance claim probability modelling. **63% F1-Score** with XGBoost optimized for heavy class imbalance and **temporal holdout validation** (no random-split data leakage -- critical when claim patterns shift over time). Class weight balancing for asymmetric error cost. Interactive **Django EDA platform** lets non-technical reviewers explore model decisions. Containerized production deployment. Pattern transfers directly to fraud-claim detection, churn prediction, and customer-risk scoring.

Python, XGBoost, CatBoost, Django, Docker, Pandas, scikit-learn | github.com/ccemozclk/Credit-Risk

Retail Demand Forecasting End-to-End MLOps Pipeline

~**12.83% RMSPE** on 6-week daily sales forecasting across 1,115 Rossmann stores using time-based validation (no temporal leakage). Object-oriented training pipeline (DataIngestion -> DataTransformation -> ModelTrainer) replacing static notebook code -- the canonical end-to-end lifecycle pattern. **Optuna hyperparameter search**, FastAPI REST inference, and Streamlit dashboard with **dynamic confidence intervals** translating ML metrics into business terms. Resilient inference layer handling unseen categorical inputs in production.

Python, LightGBM, XGBoost, Optuna, FastAPI, Streamlit, Docker | github.com/ccemozclk/Rossmann-Demand-Forecasting

EDUCATION

B.Sc. Industrial Engineering — Kırıkkale University

Sept 2016 -- Feb 2021

Quantitative degree (Mathematics, Engineering, Statistics core) with rigorous foundation in statistical analysis, operations research, and optimization. Designed MILP models with IBM ILOG CPLEX; applied AHP and TOPSIS to multi-criteria decision problems -- direct prerequisite for cost-function-aware modelling in insurance pricing and risk scoring.

CERTIFICATIONS

Associate Data Scientist Career Track (incl. PySpark, Recommendation Engines) — [DataCamp](#)

Jan -- May 2024

Data Science & Machine Learning Bootcamp (6 months, 10+ projects) — [Miuul](#)

Mar -- Sept 2022

SKILLS

Languages: Python (Advanced), SQL (Advanced), Bash

LLM Ecosystem: Hugging Face (Datasets + Transformers), SentenceTransformers, Qdrant (Vector DB), OpenAI / Llama / Gemini (exploration), Custom Embedding Training (Word2Vec, Item2Vec)

ML & Statistical: XGBoost, LightGBM, scikit-learn, h2o.ai, Statistical Modelling, A/B Testing, Class Imbalance Handling, Temporal Validation, Feature Engineering, Optuna Hyperparam Search

Deep Learning: TensorFlow, Keras, PyTorch, CNN, RNN, LSTM (Bi-LSTM custom), Transfer Learning, Grad-CAM (XAI)

MLOps & Deployment: Docker, FastAPI, Django, REST APIs, Versioned APIs, Git, Linux, End-to-End ML Lifecycle, AWS (EC2, S3)

Big Data: PySpark, Apache Spark, Databricks, Distributed Computing

Data Viz: Streamlit, matplotlib, seaborn, plotly, Django interactive EDA platforms

Databases: PostgreSQL, MSSQL, Oracle, ETL Pipelines, Pandas, NumPy, Redis

Advanced: Cost-Function Optimization, MILP, AHP/TOPSIS, OOP, System Architecture **Languages:** Turkish (Native), English (Fluent)